Introduction to the National Pollutant Release Inventory (NPRI)



October 6, 2026

Agenda of the presentation

Chapter 1- Introduction

- Overview of the NPRI Program
- History of the NPRI
- What information is reported to the NPRI?
- Which substances are reported?
- Who needs to report to the NPRI?

Chapter 2 - **Annual activities and mapping**

- NPRI reporting cycle
- Quality control of data
- Mapping data

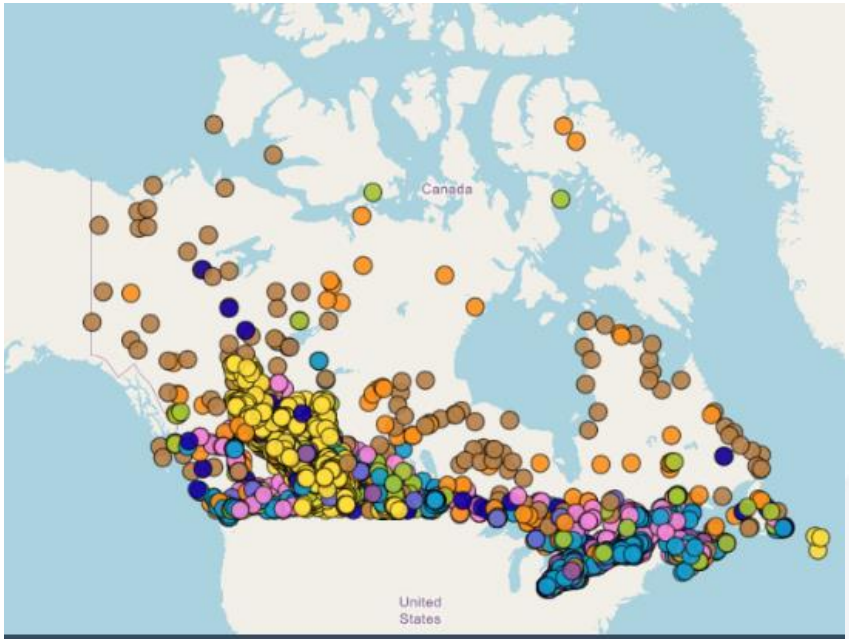
Chapter 3 - **Using NPRI and data products**

- Interpretation and use of data
- Analyzing data
- Tools for Accessing NPRI Data
- NPRI data application



Chapter 1 - overview of the NPRI program

- The NPRI is Canada's Pollutant Release and Transfer Register.
- It falls under the Canadian Environmental Protection Act, 1999 (CEPA), and is implemented by Environment and Climate Change Canada (ECCC).
- The NPRI covers releases, disposals and transfers for recycling of over 491 pollutants from over 7,000 facilities.
- The NPRI provides tools and guidance to assist facilities with reporting.



National Pollutant Release Inventory Map

History of the NPRI

- Reporting requirements were established in 1992 by a multi-stakeholder advisory committee, with representatives from federal, provincial and territorial governments, industry and environmental organization.

- Original NPRI list of substances was derived through stakeholder consultation.
  - A large measure of consensus was achieved, with several items to be discussed as the program develops.

- In 1993, the first reporting year, 178 substances were listed.

Starting point: Toxics Release Inventory substance list

- The United States' Toxics Release Inventory (TRI) substance list from 1990 was used as a starting point.

- In 1990, the Toxics Release Inventory listed about 320 substances/groups of substances that cause:

  - Cancer or other chronic human health effects
  - Significant adverse acute human health effects
  - Significant adverse environmental effects

- Substances included persistent organic pollutants, metals, ozone-depleting substances, pesticides, etc.

- A number of substances were removed from consideration – those that were not used in Canada, determined using Canada's *Domestic Substances List*.

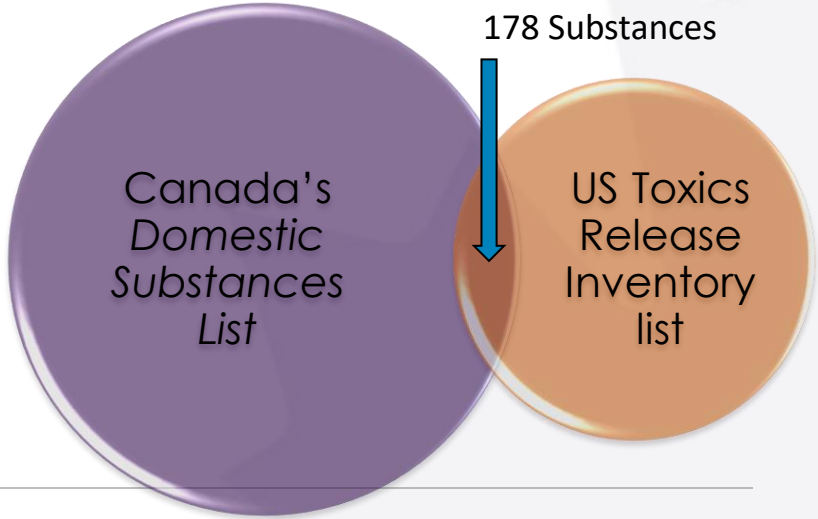
Initial NPRI candidate list

- Substances on both the Toxic Release Inventory list and the *Domestic Substances List*.

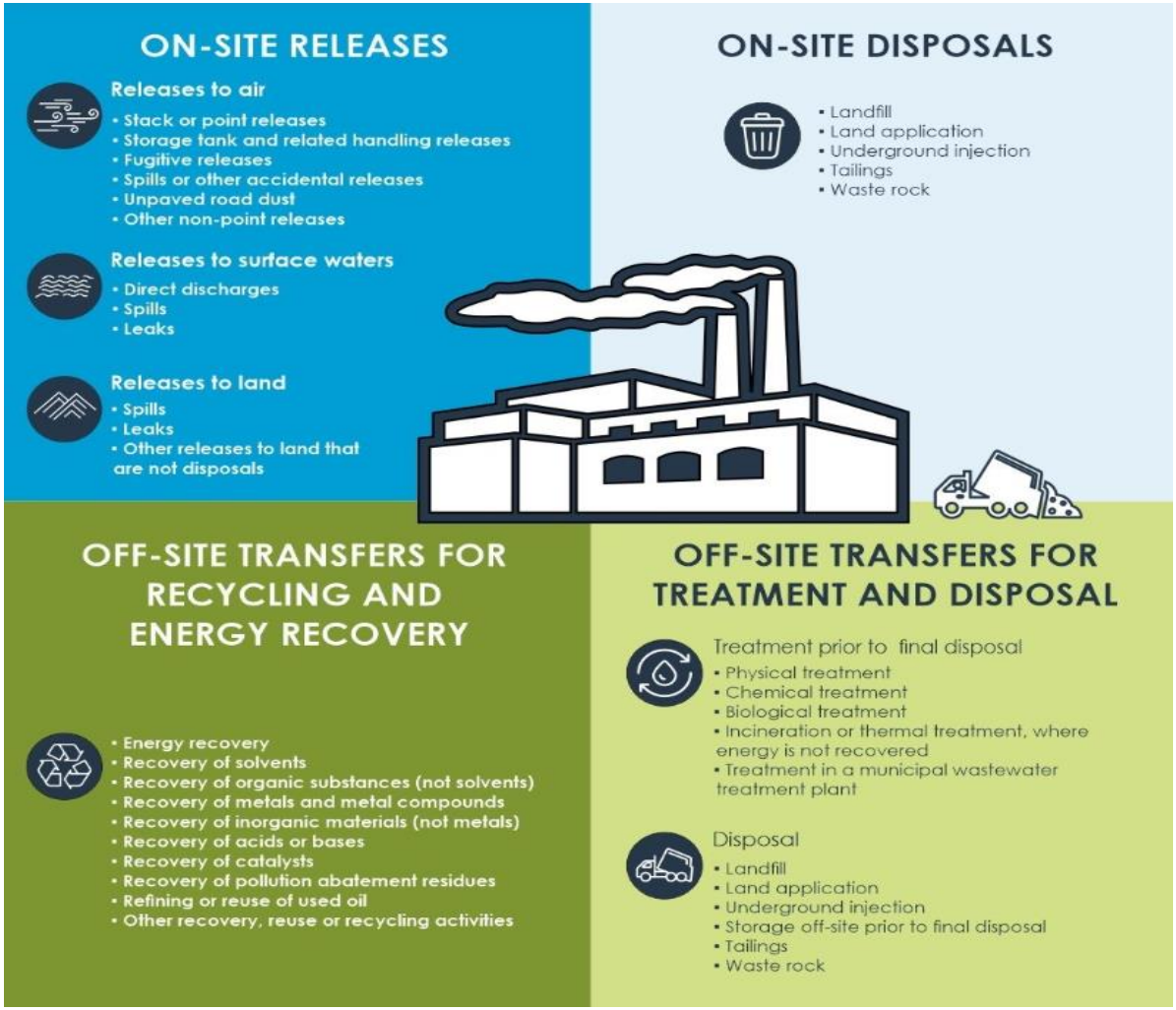
- Substances that were used in quantities smaller than 1 metric tonne per year were removed from consideration.

    - Considered a reasonable cut-off, since the original threshold was 10 metric tonnes.

- Substances also removed from consideration:
    - Pesticides
    - Ozone-depleting substances
    - Certain regulated or banned substances



What information is reported to the NPRI?



- **Pollutant Quantities**

Releases: air, surface water, land and <1 tonne in total

Disposals (on- or off-site): landfill, land application, underground injection, tailings and waste rock.

Transfers (off-site): treatment, recycling or energy recovery (including the destination of the transfer)

- **Other information**

Facility information: location, industry type, contacts

Pollution Prevention activities

Reasons for changes compared to quantities reported from previous years

And more…

Which substances are reported?

Part	Reporting criteria
Part 1A	Core Substances that must be reported if they are manufacture, processed or otherwise used, referred to as MPO for short, in amounts exceeding the reporting thresholds (≥10 tonnes at 1% concentration).
Part 1B	Substances with alternate reporting threshold that must be reported when they are MPO at lower reporting concentration or in smaller quantities.
Part 1C	These substances are per- and polyfluoroalkyl substances (PFAS). PFAS are used in many commercial applications and industrial sectors and are found in a wide range of products.
Part 2	Polycyclic Aromatic Hydrocarbons (PAHs) with a higher threshold for grouped PAHs and a smaller threshold for individual PAHs.
Part 3	Dioxins, furans and hexachlorobenzene produced by specific activities, mainly as by-producsts of industrial and combustion processes.
Part 4	Criteria Air Contaminants (CACs). These substances are produced primarily by processes such as combustion, incineration and other industrial processes and contribute to air pollution issues such as acid rains and smog.
Part 5	These are individual or speciated volatile organic compounds (VOCs) that must be reported when releases to the air in a quantity ≥ 1 tonne.

Who needs to report to the NPRI?

Reporting to the NPRI is mandatory under the *Canadian Environmental Protection Act, 1999* (CEPA). Reports are due as indicated in the Canada Gazette. Owners and operators of facilities that meet the NPRI reporting requirements published in the Canada Gazette, Part I, must report to the NPRI.

In general, facilities are required to report to the NPRI if:

- the total number of hours worked at the facility exceeded the 20,000-hour threshold (approximately 10 full-time employees)
- they are engaged in certain activities (for example: wood preservation, incineration or burning fuels in stationary combustion equipment)
- they manufacture, process, use or release NPRI substances in quantities above the reporting thresholds



What information is not reported to the NPRI?

The NPRI does not provide a comprehensive picture of all sources of pollution in Canada. However, it does paint a general portrait of pollutant releases over time.

- The NPRI does not collect data on nonpoint source emissions. The Air Pollutant Emissions Inventory (APEI) produces reports that include additional sources such as transportation and agriculture.

- Not all Canadian facilities are required to report to the NPRI (e.g., they don't meet the thresholds).

- Not all substances are listed on the NPRI. Some substances not found on the NPRI may be regulated by other ECCC programs, including the:

    - Greenhouse Gas Reporting Program (GHGRP);
    - Air Pollutant Emissions Inventory (APEI);
    - Black Carbon Emissions Inventory.

Types of facilities

Some industrial sectors:

- Oil and gas extraction, chemical pulp mills, petroleum refineries, mining, metal manufacturing, chemical manufacturing

Treatment facilities:

- Waste treatment and disposal, sewage treatment facilities

Some services facilities:

- Electric power generation, airports, cargo handling, research and development facilities, hospitals

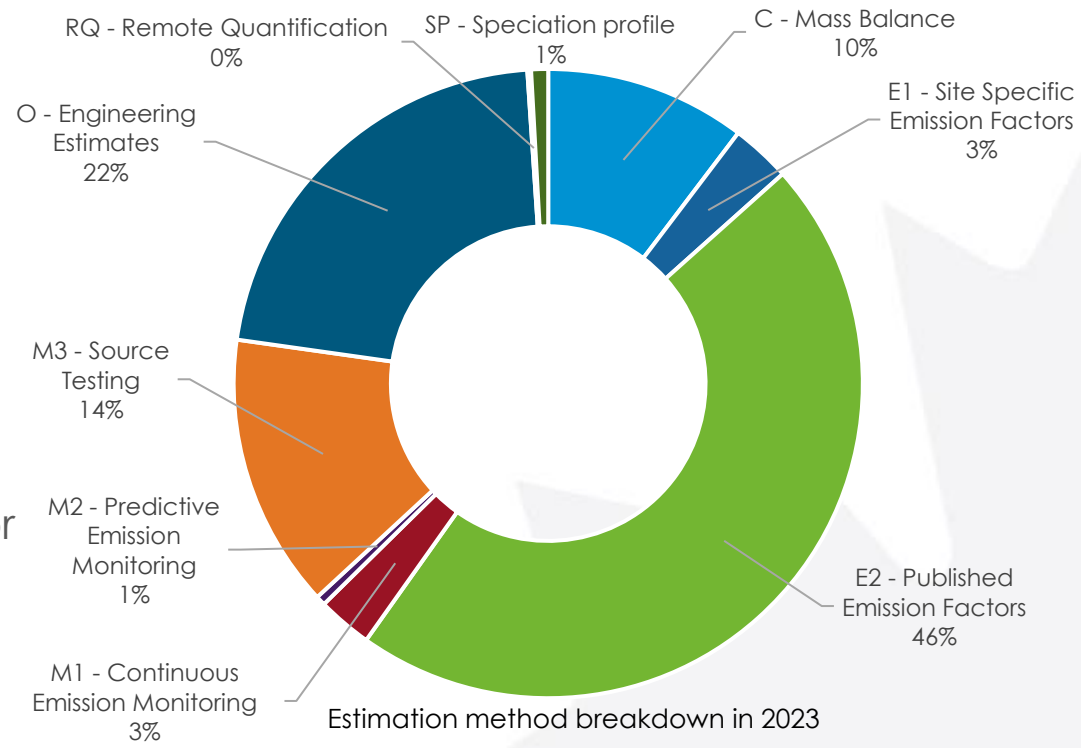
Government facilities:

- National Defence, Correctional Services

Estimation methods

- Facilities are able to choose the best available method to prepare their release estimates.
  - If the facility is required by law to measure or monitor releases, disposals or transfers for recycling, the facility must use those data to report to the NPRI

- Bases of estimate:
  - continuous emission monitoring systems
  - predictive emission monitoring
  - source testing
  - remote quantification
  - mass balance
  - site-specific emission factor
  - published emission factor
  - speciation profile
  - engineering estimates



Estimation method breakdown in 2023

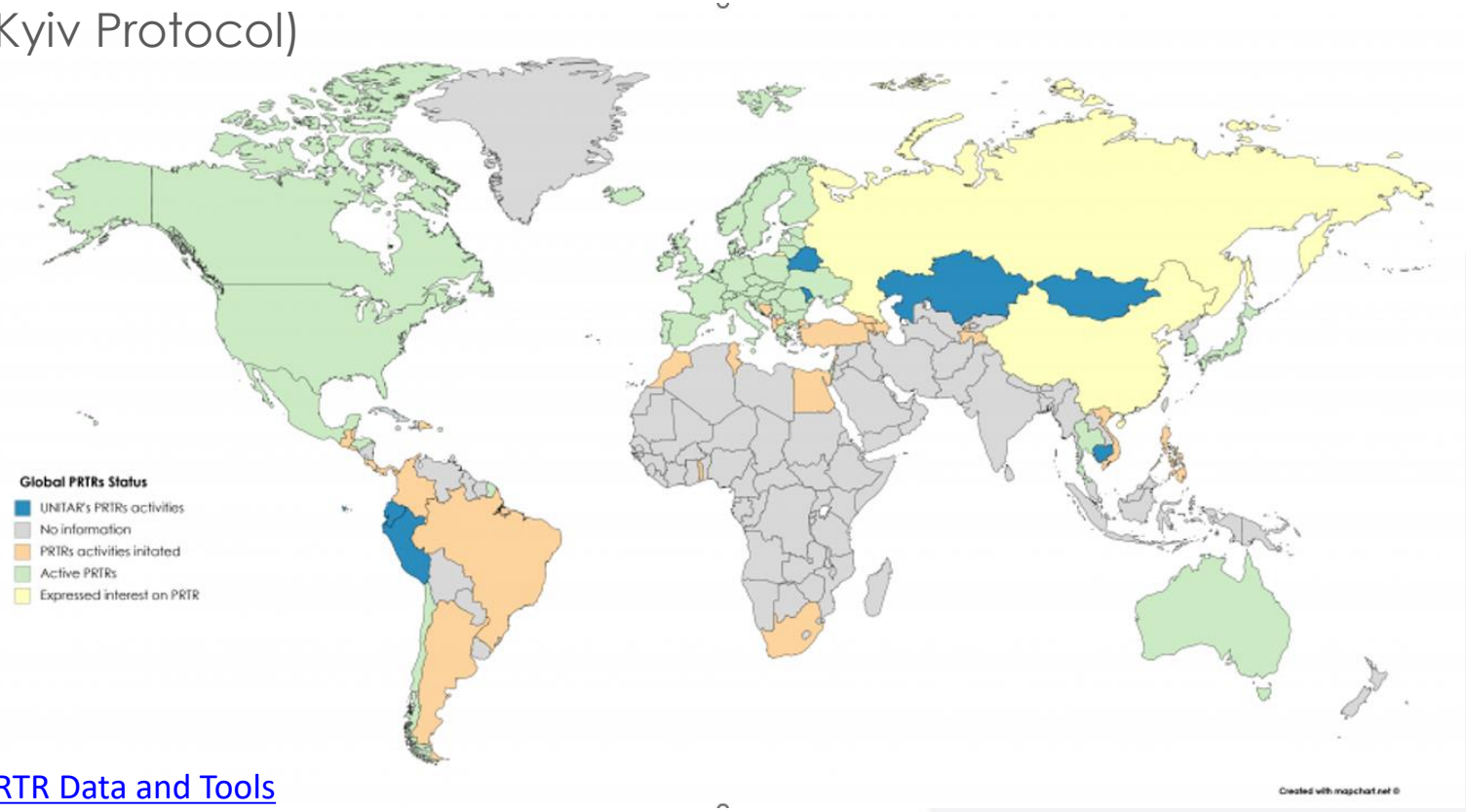
Other inventories

Program	Overview	For more information…
ECCC Greenhouse Gas Reporting Program (GHGRP)	• 27 listed GHGs • 10 kt CO2 eq. annual threshold and facilities engaged in specific activities • 1732 facilities reported in 2021	https://www.canada.ca/en/environment-climate-change/services/climate-change/greenhouse-gas-emissions/facility-reporting.html
ECCC Air Pollutant Emission Inventory (APEI)	• 17 listed substances and groups of substances • NPRI data is used for various industrial sectors including ore and mineral industries, as well as manufacturing • NPRI data is not used for sectors such as transportation and agriculture	https://www.canada.ca/en/environment-climate-change/services/pollutants/air-emissions-inventory-overview.html
Black Carbon Inventory	• Black carbon emitted in PM2.5 resulting from combustion • APEI/NPRI data used for various sectors	https://www.canada.ca/en/environment-climate-change/services/pollutants/black-carbon-emissions-inventory.html

Pollutant inventories around the world

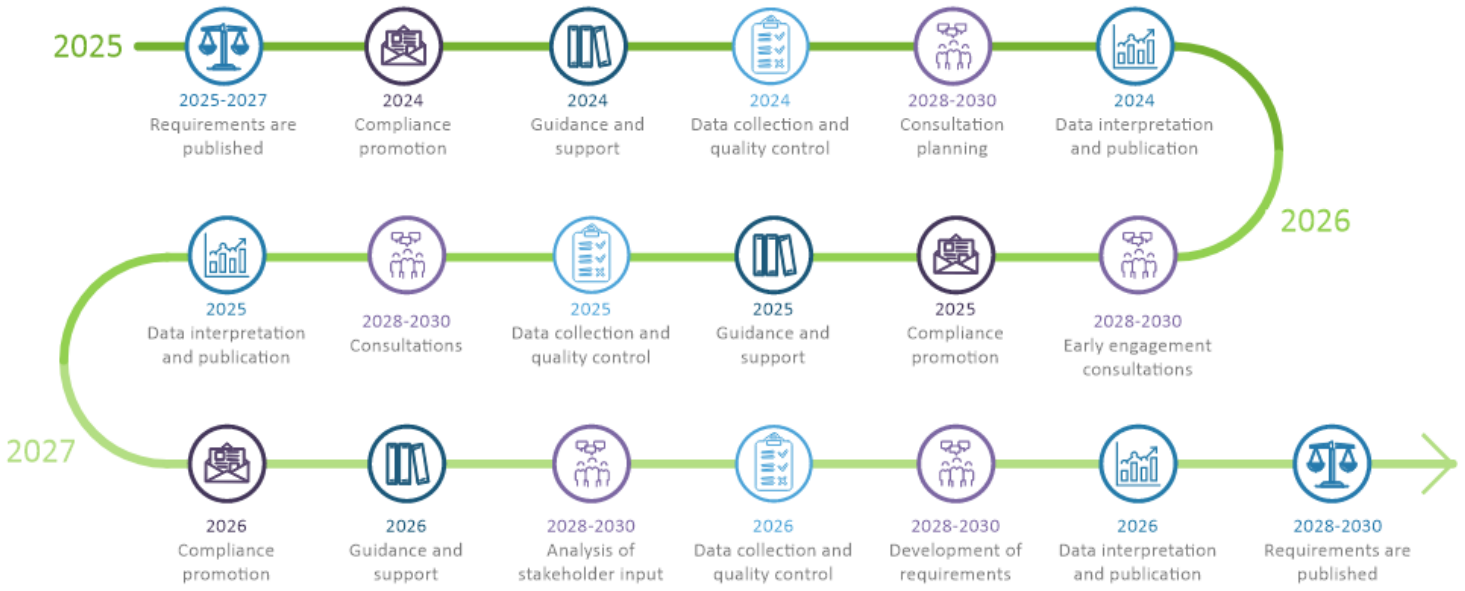
Canada participates in international partnerships to advance global PRTRs development and help countries establish their own programs

- UNECE - (Kyiv Protocol)
- OECD
- UNITAR
- CEC



International PRTR Data and Tools

Chapter 2 - The NPRI annual reporting cycle



Publication of reporting requirements
Reporting requirements are published in the Canada Gazette Part I every third year

Compliance promotion
Companies that may need to report are identified and made aware of reporting requirements each year

Guidance and support
Guides and tools for reporting are developed and published. Helpdesk responds to enquiries from reporting facilities each year

Data collection and quality control
Updates are made to the online reporting system. Reports are submitted by June 1st each year. Submitted data are reviewed and corrected

Consultation and development of reporting requirements
Future requirements are developed in consultation with stakeholders and the public over a three year period

Data interpretation and publication
Summary reports are prepared. Reports and data are published in December each year

Consultations

- A consultative process was used to design the NPRI and is used for the consideration of changes.

- Key components:
  - Established a standing multi-stakeholder group;
  - Clear expectations provided at the outset;
  - Criteria for evaluating changes; and
  - Final decision on changes remains with ECCC.



Making changes to NPRI reporting requirements

The NPRI substance list, reporting requirements, and supplementary information are reviewed regularly. Anyone can submit a request to review NPRI reporting requirements.

1. For a substance:
    - Is it manufactured, processed or otherwise used in Canada?
    - Is it of health and/or environmental concern?
    - Do facilities contribute significant releases of the substance?
    - Does its inclusion support one or more of the objectives of the NPRI?

2. For modification of reporting thresholds:
    - Science-based justification;
    - Capability of facilities to provide the required information;
    - Evidence of the significance of releases.

Compliance promotion

- The NPRI sends reminder emails to facilities at the launch of reporting, and near the reporting deadline;
- Targets sectoral activities based on identified priorities to promote compliance and inform potentially non-reporting facilities of the reporting requirements, completeness and accuracy of submitted reports;
- Enforcement action is sometimes necessary.



Advice and support: Guidelines for reporting to the NPRI

NPRI Report Types

- Inventory Report
- Does Not Meet Criteria (DNMC) Report
- Sale, Closure or Purchase Report

- If a facility reported the previous year, but no longer meets the reporting requirements, they must submit a DNMC
- Facilities are required to submit a report if the facility has been sold, closed or purchased
- Both these types of reports help with QC and compliance promotion since it helps us track which facilities belong to who and whether we need to follow up

Guidance and support: tools to help facilities calculate emissions

- The NPRI provides different tools to help facilities estimate their releases, disposals and transfers

- Sector specific tools include:
  - e.g. aluminium production, asphalt plants, beer brewing, grain handling, wood preservation

- General tools include:
  - Fuel combustion calculators
    - Diesel, natural gas, gasoline, fuel oil
  - Fugitive emissions calculators
    - Dust from unpaved roads
    - Abrasive blasting
    - Wet cooling towers

Data collection: how to report to the NPRI?

Facilities that meet the NPRI's reporting requirements must complete a comprehensive report on their pollutant releases, disposals and transfers using the online reporting system which is part of Environment and Climate Change Canada's Single Window (SW).

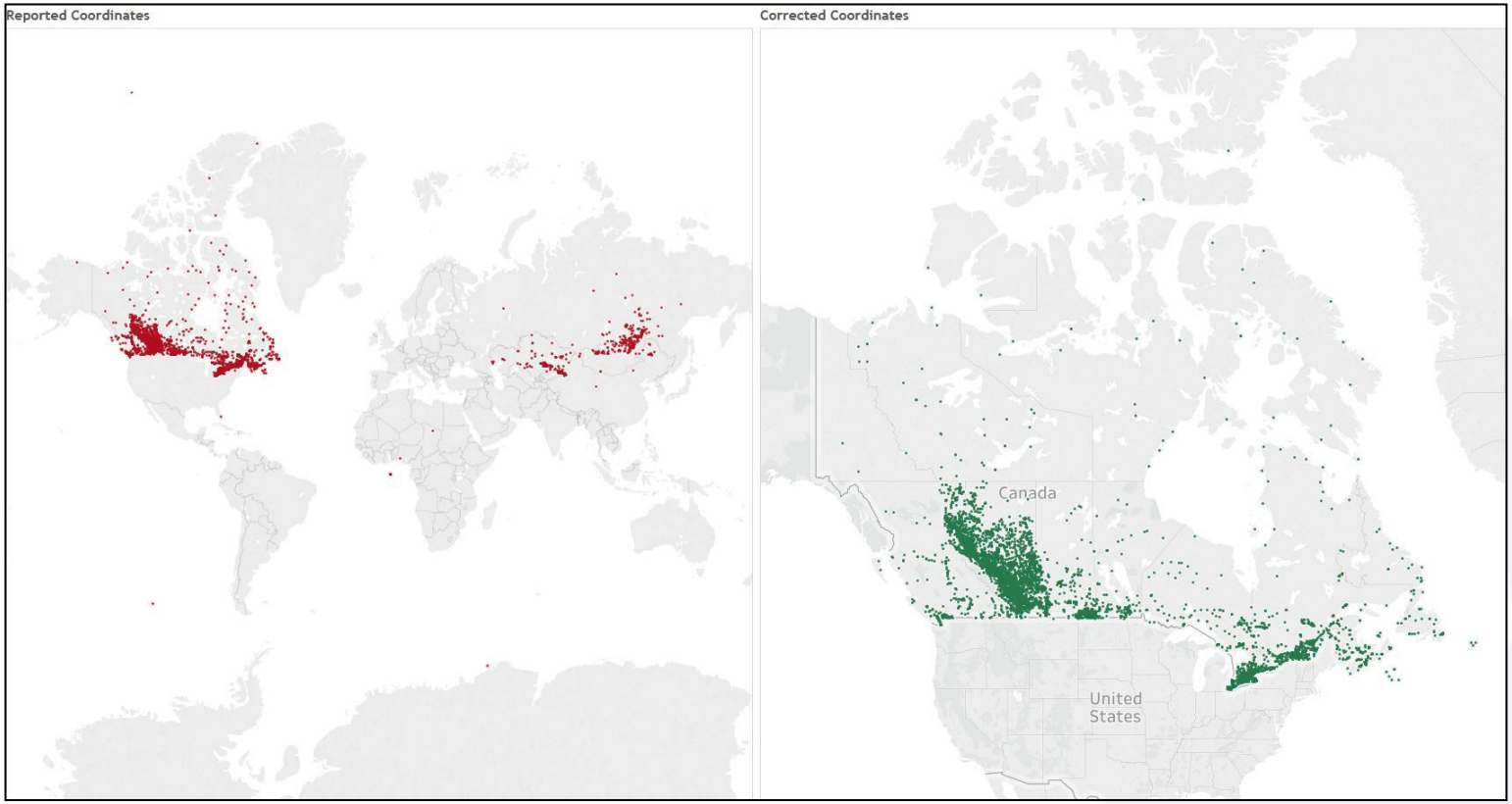
- National Pollutant Release Inventory
- Greenhouse Gas Reporting
- Mine Effluent Reporting System
- Effluent Regulatory Reporting Information System
- Canadian Notification and Movement Tracking System
- Pollution Prevention Planning Notices
- Environmental Emergency Regulations Reporting System

Quality control of data

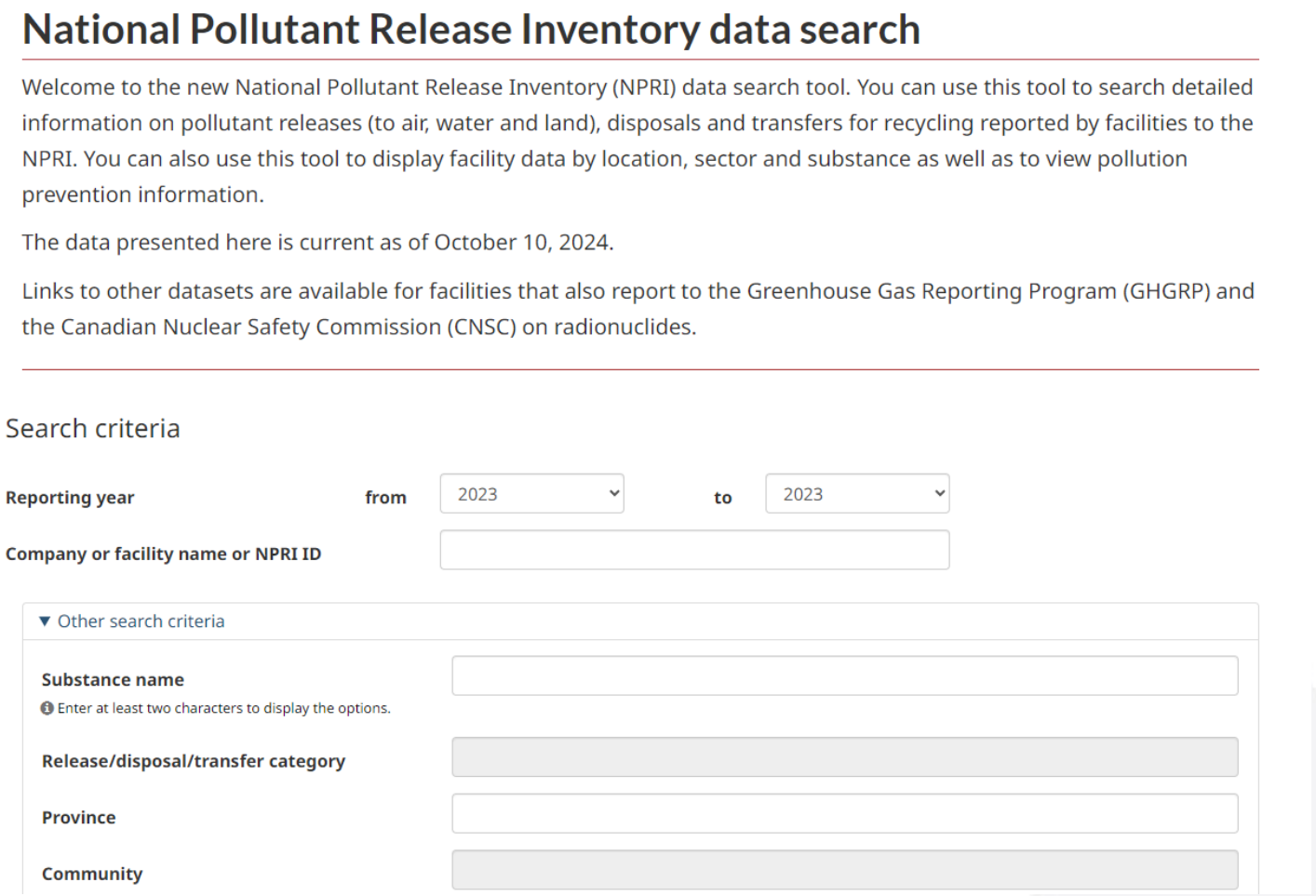
- Data assurance and quality control are performed by:
    - NPRI employees;
    - Environnement and Climate Change Canada sector experts;
    - Industry associations
- Facilities are regularly encouraged to verify their own data.
- Examples of common reporting errors observed during quality control activities:
    - Using inappropriate reporting units and decimal errors:
    - Omitting a substance or the source of a release;
    - Using incorrect reference data or assumptions for calculations;
    - Forgetting the (-) sign in the facility's geographical coordinates.

Quality control of data

- Incorrect latitude and longitude coordinates can lead to interesting facility locations



Data publication: online data search



National Pollutant Release Inventory data search

Welcome to the new National Pollutant Release Inventory (NPRI) data search tool. You can use this tool to search detailed information on pollutant releases (to air, water and land), disposals and transfers for recycling reported by facilities to the NPRI. You can also use this tool to display facility data by location, sector and substance as well as to view pollution prevention information.

The data presented here is current as of October 10, 2024.

Links to other datasets are available for facilities that also report to the Greenhouse Gas Reporting Program (GHGRP) and the Canadian Nuclear Safety Commission (CNSC) on radionuclides.

Search criteria

Reporting year **from** 2023 **to** 2023

Company or facility name or NPRI ID

▼ Other search criteria

Substance name

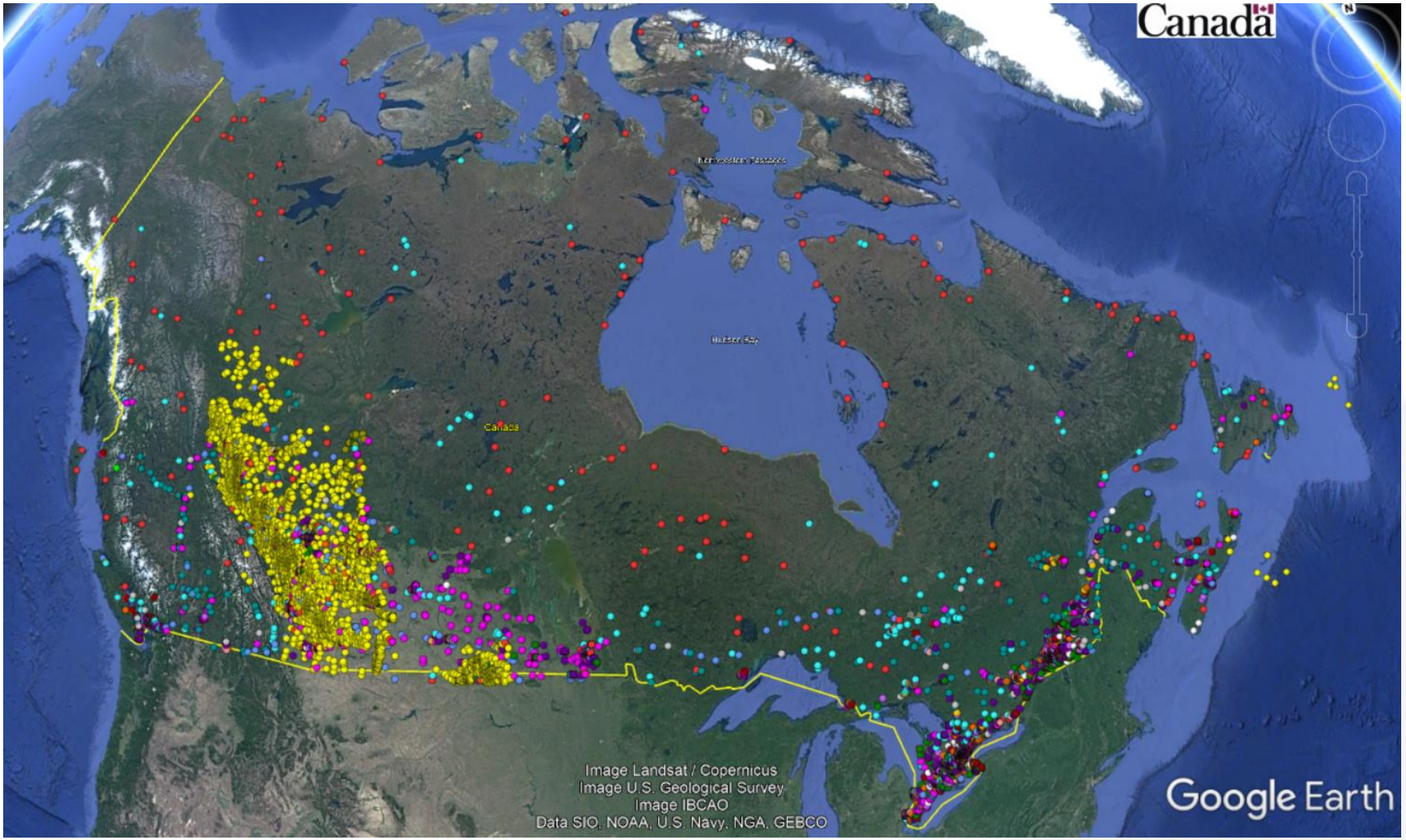
Enter at least two characters to display the options.

Release/disposal/transfer category

Province

Community

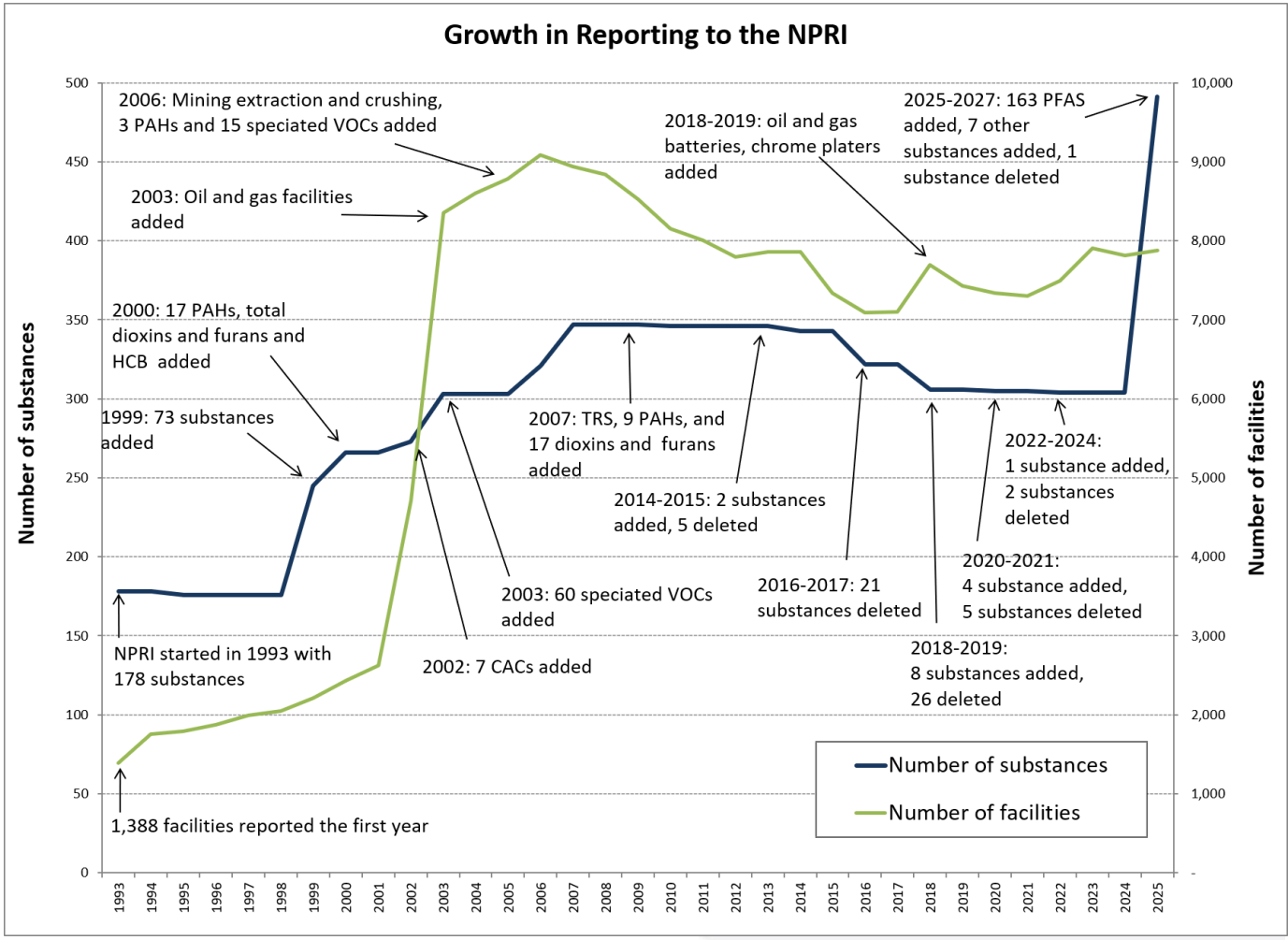
Data publication: mapping data



Chapter 3 : Using NPRI data

Since the program was created, many substances have been modified, added or deleted.

It is important to always consult the history of reporting requirements before analyzing or interpreting NPRI data over time, to avoid false conclusions.



Interpretation and use of data

The guide for using and interpreting NPRI Data is a resource available on the NPRI website. You can use it to identify potential sources of error and factors to consider before interpreting datasets.

Avoiding double-counting (an example using TRS):

Substance	Substances in TRS	Releases to air (tonnes)
Total reduced sulphur		15 523
	Carbonyl sulphide	5 159
	Carbon disulphide	4 458
	Hydrogen sulphide	2 707
	Other TRS substances	3 199
Total		**15 523**

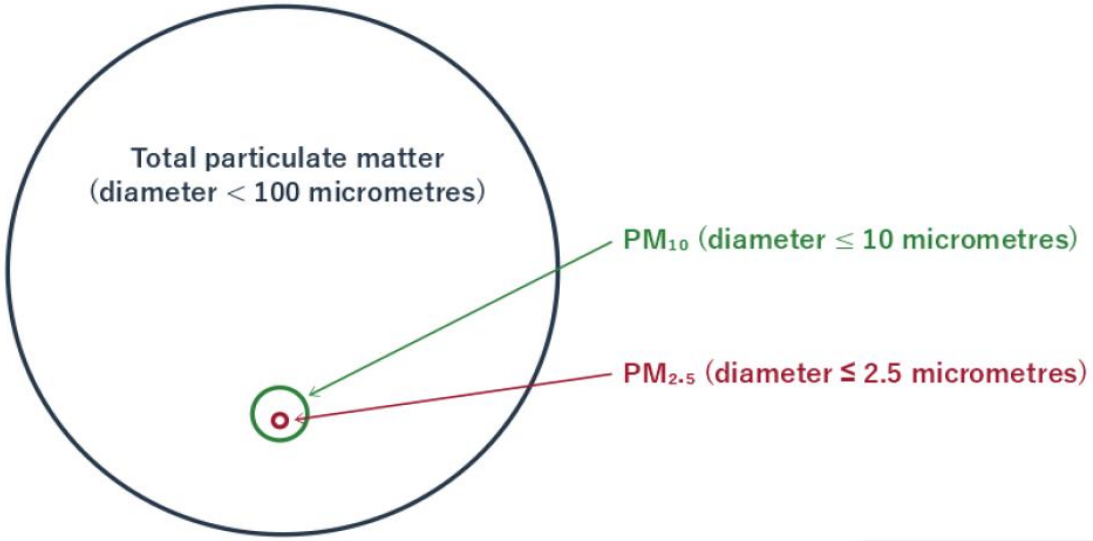
Substance	Releases to air (tonnes)
Total reduced sulphur	15 523
Carbonyl sulphide	5 159
Carbon disulphide	4 458
Hydrogen sulphide	2 707
Total	**27 847**

Interpretation and use of data: PM

- Reporting thresholds for particulate matter:

Size fraction	Threshold
$PM_{2.5}$	0.3 tonnes
PM_{10}	0.5 tonnes
Total PM (TPM)	20 tonnes

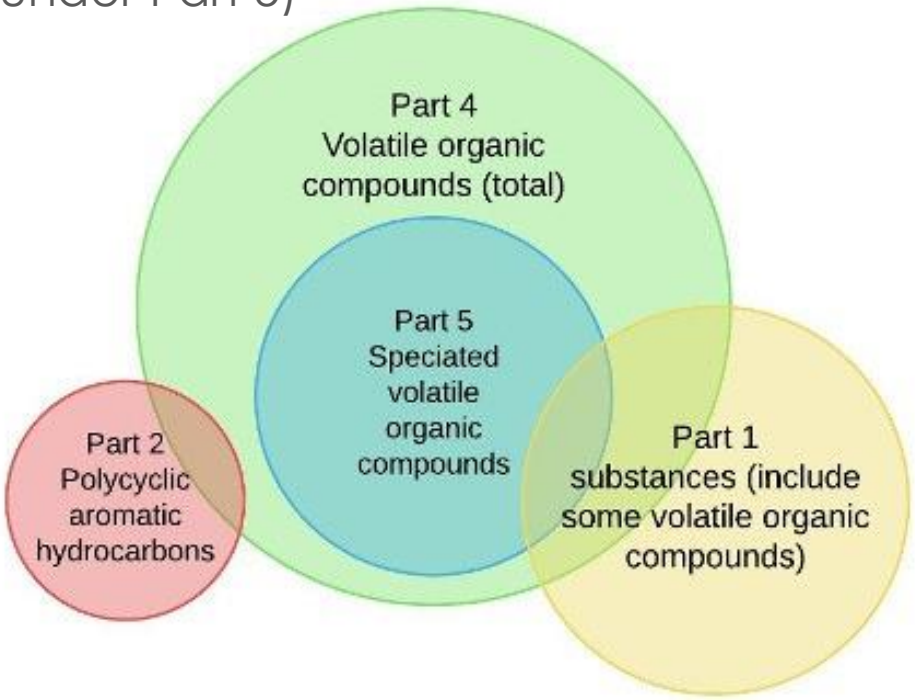
- Can meet the threshold(s) for one or two of the smaller fractions without necessarily meeting the threshold(s) for larger fraction(s)



Interpretation and use of data: VOCs

The NPRI has over 100 VOCs listed under different reporting criteria. Facilities may have to report them as:

- a total of all volatile organic compounds, under Part 4
- individual compounds (under Parts 1 and 2, and speciated compounds under Part 5)



Analyzing data

- Various dimensions of analysis possible: geolocation, industry, substance, facility, company, environmental compartment, pollution prevention data, timeline, etc.
- Contextual data available: reasons for change, comments, facility activities, etc.

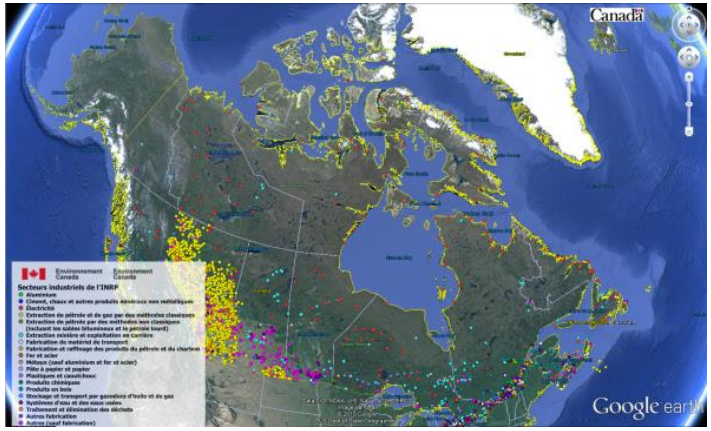
NPRI Information products

- **Data Highlights:** A high-level annual summary of the most recently reported data
- **Overview Series:** Interactive maps on targeted sectors and substances of interest
- **Data Integration Series:** Interactive maps designed to demonstrate linkages with other environmental datasets

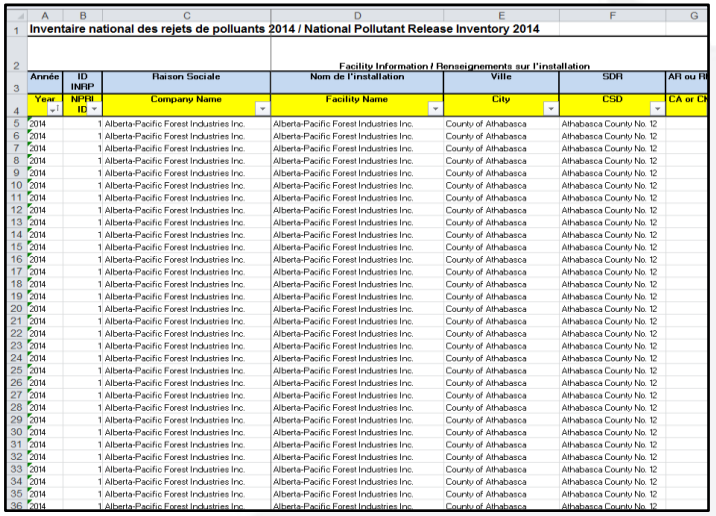
Tools for accessing NPRI data

[NPRI Online Query Site]
Quick, easy to find information on a particular facility.

[NPRI Map Layers for Use in Google Earth™]
Explore NPRI data in an intuitive and visual way.



[MS Excel or CSV Files – Tabular Data]
Most common fields in a simplified spreadsheet format.



Inventaire national des rejets de polluants 2014 / National Pollutant Release Inventory 2014

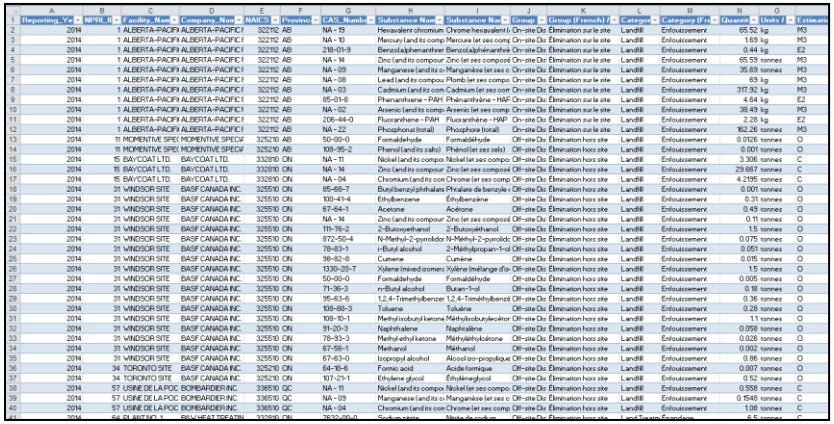
Facility Information / Renseignements sur l'installation

Année / Year	ID INRP / NPRI ID	Raison Sociale / Company Name	Nom de l'installation / Facility Name	Ville / City	SDR / CSD	AR ou FI / CA or CI
2014	1	Alberta-Pacific Forest Industries Inc.	Alberta-Pacific Forest Industries Inc.	County of Athabasca	Athabasca County No. 12	
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Tools for accessing NPRI data

MS Excel or CSV files – Bulk Data

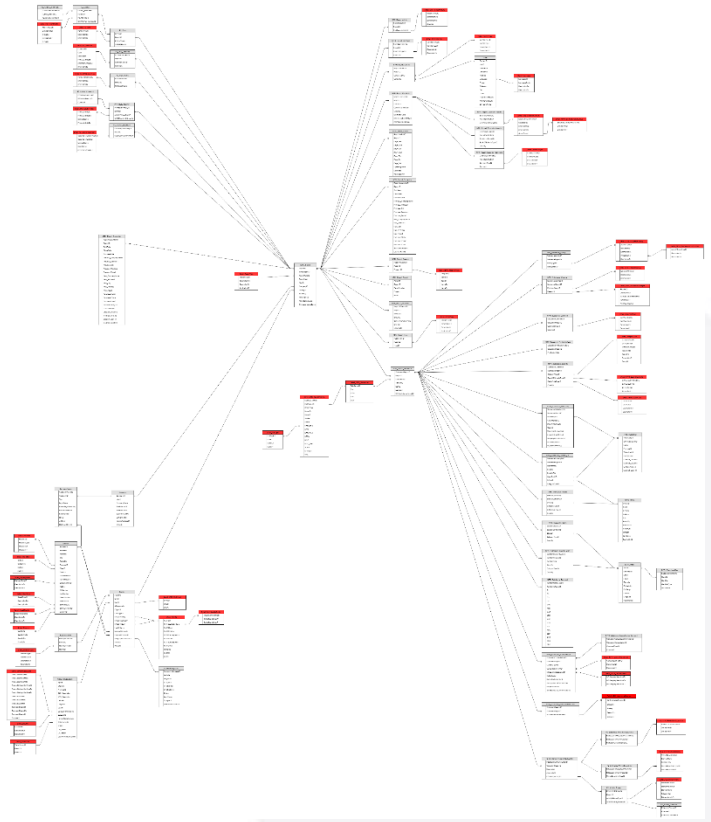
Analyze grouped data by sets for releases, disposals and transfers for all reporting years since 1993.



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	
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2	2014	1	ALBERTA-PACIFI	ALBERTA-PACIFIC I	322112	AB	NA - 13	Hexavalent chromium	Chrome hexavalent ((	On-site Dis	Élimination sur le site	Landfill	Enfouissement	65.52	kg	M3
3	2014	1	ALBERTA-PACIFI	ALBERTA-PACIFIC I	322112	AB	NA - 10	Mercury (and its comp	Mercure (et ses comp	On-site Dis	Élimination sur le site	Landfill	Enfouissement	1.69	kg	M3
4	2014	1	ALBERTA-PACIFI	ALBERTA-PACIFIC I	322112	AB	218-01-9	Benzo(a)phenanthren	Benzo(a)phénanthrène	On-site Dis	Élimination sur le site	Landfill	Enfouissement	0.44	kg	E2
5	2014	1	ALBERTA-PACIFI	ALBERTA-PACIFIC I	322112	AB	NA - 14	Zinc (and its compoun	Zinc (et ses composé	On-site Dis	Élimination sur le site	Landfill	Enfouissement	65.59	tonnes	M3
6	2014	1	ALBERTA-PACIFI	ALBERTA-PACIFIC I	322112	AB	NA - 09	Manganese (and its co	Manganèse (et ses c	On-site Dis	Élimination sur le site	Landfill	Enfouissement	35.69	tonnes	M3
7	2014	1	ALBERTA-PACIFI	ALBERTA-PACIFIC I	322112	AB	NA - 08	Lead (and its compou	Plomb (et ses compo	On-site Dis	Élimination sur le site	Landfill	Enfouissement	69	kg	M3
8	2014	1	ALBERTA-PACIFI	ALBERTA-PACIFIC I	322112	AB	NA - 03	Cadmium (and its com	Cadmium (et ses com	On-site Dis	Élimination sur le site	Landfill	Enfouissement	317.92	kg	M3
9	2014	1	ALBERTA-PACIFI	ALBERTA-PACIFIC I	322112	AB	85-01-8	Phenanthrene - PAH	Phénanthrène - HAP	On-site Dis	Élimination sur le site	Landfill	Enfouissement	4.64	kg	E2
10	2014	1	ALBERTA-PACIFI	ALBERTA-PACIFIC I	322112	AB	NA - 02	Arsenic (and its comp	Arsenic (et ses comp	On-site Dis	Élimination sur le site	Landfill	Enfouissement	38.49	kg	M3
11	2014	1	ALBERTA-PACIFI	ALBERTA-PACIFIC I	322112	AB	206-44-0	Fluoranthene - PAH	Fluoranthène - HAP	On-site Dis	Élimination sur le site	Landfill	Enfouissement	2.28	kg	E2
12	2014	1	ALBERTA-PACIFI	ALBERTA-PACIFIC I	322112	AB	NA - 22	Phosphorus (total)	Phosphore (total)	On-site Dis	Élimination sur le site	Landfill	Enfouissement	162.26	tonnes	M3
13	2014	11	MOMENTIVE SPEC	MOMENTIVE SPECIA	325210	AB	50-00-0	Formaldehyde	Formaldéhyde	Off-site Dis	Élimination hors site	Landfill	Enfouissement	0.0126	tonnes	O
14	2014	11	MOMENTIVE SPEC	MOMENTIVE SPECIA	325210	AB	108-95-2	Phenol (and its salts)	Phénol (et ses sels)	Off-site Dis	Élimination hors site	Landfill	Enfouissement	0.001	tonnes	O
15	2014	15	BAYCOAT LTD.	BAYCOAT LTD.	332810	ON	NA - 11	Nickel (and its compou	Nickel (et ses compo	Off-site Dis	Élimination hors site	Landfill	Enfouissement	3.306	tonnes	C
16	2014	15	BAYCOAT LTD.	BAYCOAT LTD.	332810	ON	NA - 14	Zinc (and its compoun	Zinc (et ses composé	Off-site Dis	Élimination hors site	Landfill	Enfouissement	29.887	tonnes	C
17	2014	15	BAYCOAT LTD.	BAYCOAT LTD.	332810	ON	NA - 04	Chromium (and its com	Chrome (et ses comp	Off-site Dis	Élimination hors site	Landfill	Enfouissement	4.2195	tonnes	C
18	2014	31	WINDSOR SITE	BASF CANADA INC.	325510	ON	85-68-7	Butylbenzylphthalate	Phtalate de benzyle e	Off-site Dis	Élimination hors site	Landfill	Enfouissement	0.001	tonnes	O
19	2014	31	WINDSOR SITE	BASF CANADA INC.	325510	ON	100-41-4	Ethylbenzene	Éthylbenzène	Off-site Dis	Élimination hors site	Landfill	Enfouissement	0.31	tonnes	O
20	2014	31	WINDSOR SITE	BASF CANADA INC.	325510	ON	67-64-1	Acetone	Acétone	Off-site Dis	Élimination hors site	Landfill	Enfouissement	0.43	tonnes	O
21	2014	31	WINDSOR SITE	BASF CANADA INC.	325510	ON	NA - 14	Zinc (and its compoun	Zinc (et ses composé	Off-site Dis	Élimination hors site	Landfill	Enfouissement	0.11	tonnes	O
22	2014	31	WINDSOR SITE	BASF CANADA INC.	325510	ON	111-76-2	2-Butoxyethanol	2-Butoxyéthanol	Off-site Dis	Élimination hors site	Landfill	Enfouissement	1.5	tonnes	O
23	2014	31	WINDSOR SITE	BASF CANADA INC.	325510	ON	872-50-4	N-Methyl-2-pyrrolidor	N-Méthyl-2-pyrrolidc	Off-site Dis	Élimination hors site	Landfill	Enfouissement	0.075	tonnes	O
24	2014	31	WINDSOR SITE	BASF CANADA INC.	325510	ON	78-83-1	i-Butyl alcohol	2-Méthylpropan-1-ol	Off-site Dis	Élimination hors site	Landfill	Enfouissement	0.051	tonnes	O
25	2014	31	WINDSOR SITE	BASF CANADA INC.	325510	ON	98-82-8	Cumene	Cumène	Off-site Dis	Élimination hors site	Landfill	Enfouissement	0.015	tonnes	O
26	2014	31	WINDSOR SITE	BASF CANADA INC.	325510	ON	1330-20-7	Xylene (mixed isomers	Xylène (mélange d'is	Off-site Dis	Élimination hors site	Landfill	Enfouissement	1.5	tonnes	O
27	2014	31	WINDSOR SITE	BASF CANADA INC.	325510	ON	50-00-0	Formaldehyde	Formaldéhyde	Off-site Dis	Élimination hors site	Landfill	Enfouissement	0.005	tonnes	O
28	2014	31	WINDSOR SITE	BASF CANADA INC.	325510	ON	71-36-3	n-Butyl alcohol	Butan-1-ol	Off-site Dis	Élimination hors site	Landfill	Enfouissement	0.18	tonnes	O
29	2014	31	WINDSOR SITE	BASF CANADA INC.	325510	ON	95-63-6	1,2,4-Trimethylbenzer	1,2,4-Triméthylbenzè	Off-site Dis	Élimination hors site	Landfill	Enfouissement	0.36	tonnes	O
30	2014	31	WINDSOR SITE	BASF CANADA INC.	325510	ON	108-88-3	Toluene	Toluène	Off-site Dis	Élimination hors site	Landfill	Enfouissement	0.28	tonnes	O
31	2014	31	WINDSOR SITE	BASF CANADA INC.	325510	ON	108-10-1	Methyl isobutyl ketone	Méthylisobutylcétone	Off-site Dis	Élimination hors site	Landfill	Enfouissement	1.1	tonnes	O
32	2014	31	WINDSOR SITE	BASF CANADA INC.	325510	ON	91-20-3	Naphthalene	Naphtalène	Off-site Dis	Élimination hors site	Landfill	Enfouissement	0.058	tonnes	O
33	2014	31	WINDSOR SITE	BASF CANADA INC.	325510	ON	78-93-3	Methyl ethyl ketone	Méthyléthylcétone	Off-site Dis	Élimination hors site	Landfill	Enfouissement	0.026	tonnes	O
34	2014	31	WINDSOR SITE	BASF CANADA INC.	325510	ON	67-56-1	Methanol	Méthanol	Off-site Dis	Élimination hors site	Landfill	Enfouissement	0.002	tonnes	O
35	2014	31	WINDSOR SITE	BASF CANADA INC.	325510	ON	67-63-0	Isopropyl alcohol	Alcool iso-propylique	Off-site Dis	Élimination hors site	Landfill	Enfouissement	0.86	tonnes	O
36	2014	34	TORONTO SITE	BASF CANADA INC.	325210	ON	64-18-6	Formic acid	Acide formique	Off-site Dis	Élimination hors site	Landfill	Enfouissement	0.007	tonnes	O
37	2014	34	TORONTO SITE	BASF CANADA INC.	325210	ON	107-21-1	Ethylene glycol	Éthylèneglycol	Off-site Dis	Élimination hors site	Landfill	Enfouissement	0.52	tonnes	O
38	2014	57	USINE DE LA POC	BOMBARDIER INC.	336510	QC	NA - 11	Nickel (and its compou	Nickel (et ses compo	Off-site Dis	Élimination hors site	Landfill	Enfouissement	0.558	tonnes	C
39	2014	57	USINE DE LA POC	BOMBARDIER INC.	336510	QC	NA - 09	Manganese (and its co	Manganèse (et ses c	Off-site Dis	Élimination hors site	Landfill	Enfouissement	0.1548	tonnes	C
40	2014	57	USINE DE LA POC	BOMBARDIER INC.	336510	QC	NA - 04	Chromium (and its com	Chrome (et ses comp	Off-site Dis	Élimination hors site	Landfill	Enfouissement	1.08	tonnes	C
41	2014	64	PLASTIQUE 1	PLASTIQUE 1	332811	ON	7439-92-1	Lead	Plomb	Off-site Dis	Élimination hors site	Land Treatm	Enfouissement	0.5	tonnes	C

NPRI Database

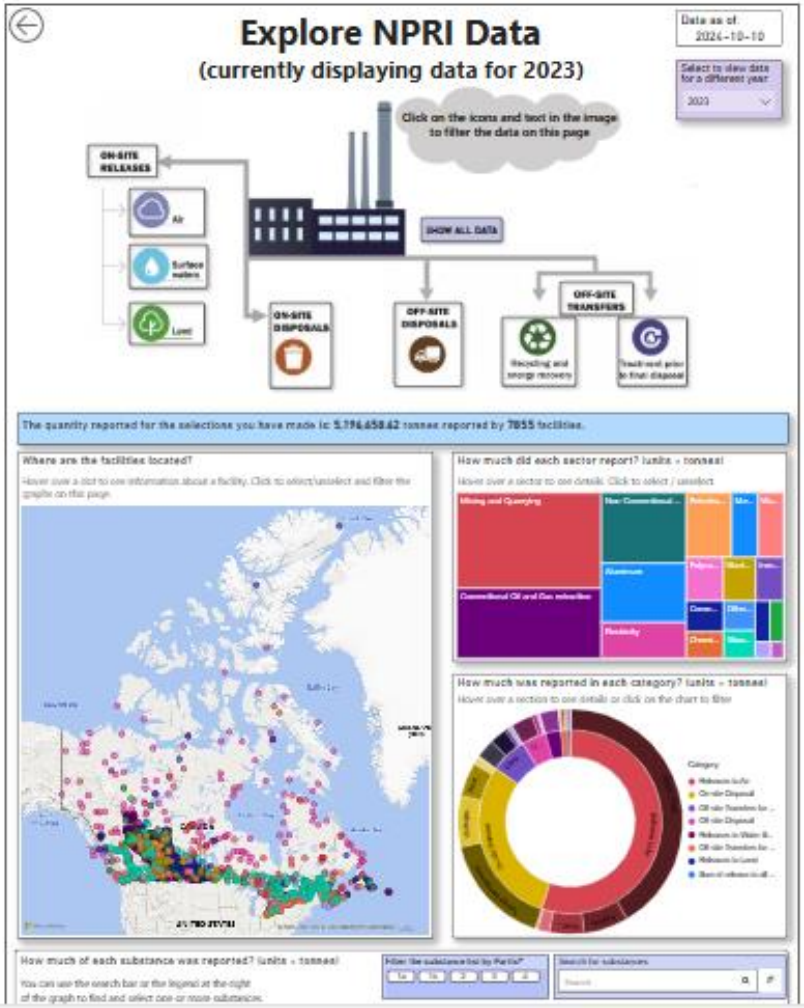
All data for all years. Useful for custom analysis for "technical" users.



Tools for accessing NPRI data

[Power BI Dashboard]

The NPRI Dashboard allows you to search data from 2013 to 2023 in visual representations such as interactive maps, graphs and tables. The search is by pollutant, facility, location or industrial sector, and provides information on pollution prevention activities.



Uses of NPRI data

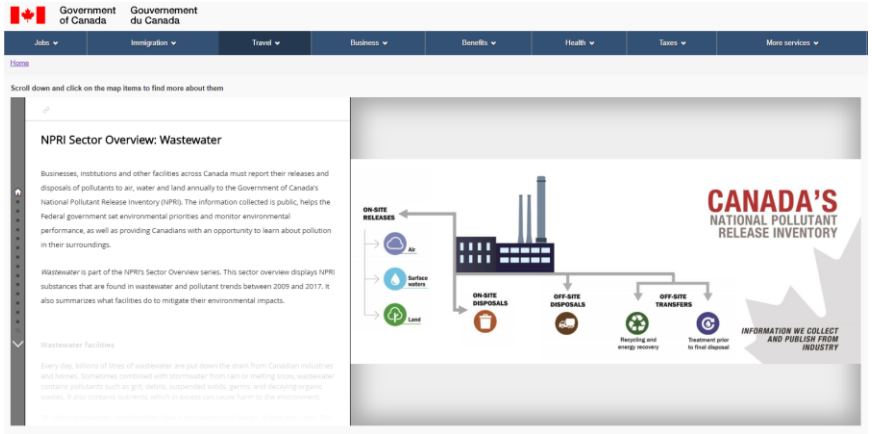
- To identify pollution prevention priorities;
- To support the assessment and management of risks from chemicals, e.g. under the Chemicals Management Plan (CMP), and air quality modelling;
- To complement monitoring data from other programs, e.g. water and air quality monitoring.
- To encourage actions to reduce releases of pollutants into the environment;
- To improve public understanding and support research on pollutants;
- To support annual international reporting.

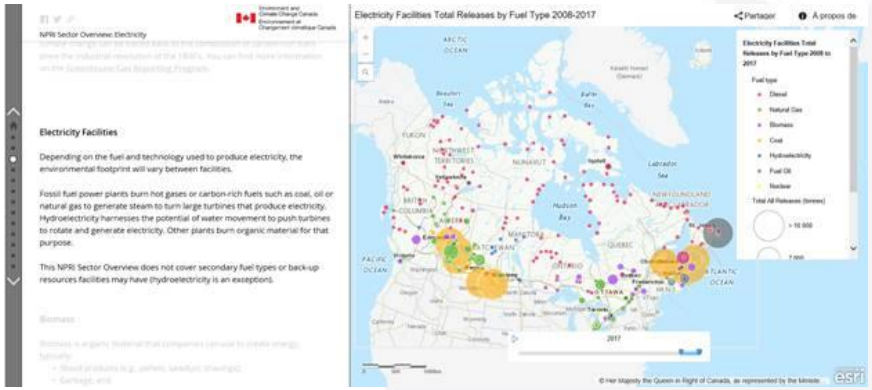
Public "right-to-know" is a fundamental underlying principle of the NPRI.

NPRI reports

- Each overview provides an overview of a specific industry, while promoting NPRI data and innovation in environmental improvements:
  - Sectors
  - Substances
  - Regional overviews
  - Data integration

- Analyzes trends in emissions over time





NPRI Academic Challenge

- Conduct innovative research projects using National Pollutant Release Inventory (NPRI) data
- Learn how to use the NPRI database and select a research topic of common interest to ECCC and the students
- Gain hands-on experience using and analyzing real-world pollution data, gain practical experience researching a science project from start to finish and contribute to the communication of science in Canada.

Examples of past projects

-Dalhousie University (2021): Human Health Implications of Mercury Exposure in Nova Scotia

-Université de Sherbrooke (2020): Case study on the pulp and paper industry in Canada, and trends in NPRI data over time

-Université du Québec en Abitibi-Témiscamingue (2021): Analysis of critical and strategic metals releases in Quebec and Canada: opportunities for the enhancement of metals in industrial discharges.

Academic Challenge

Group	Sector
1	Natural Resource Extraction (except coal)
2	Coal and lignite mining
3	Electricity Generation
4	Automotive & Transportation Manufacturing
5	Paper manufacturing
6	Wood product manufacturing
7	Petroleum processing
8	Chemical manufacturing
9	Metal Production & Processing
10	Waste Management
11	Waste Water Treatment

Project leads:

- How have pollutant releases changed over the years across different countries in each industry?
- How do facilities compare in terms of pollutants across various release mediums, disposals and transfers?
- Are there statistically significant differences between Canadian and German industrial releases between facilities? Over time?
- What might explain these differences?
- Which locations consistently show high pollution levels, and what interventions might be effective?

Thank you!
Questions?

NPRI and Substance Information Division
Science Reporting and Assessment Directorate
Science and Technology Branch
Environment and Climate Change Canada

https://canada.ca/NPRI



Academic Challenge

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- Summary
- Introduction
  - Environmental and health impacts
  - Processes that produce dioxins
- Methods
  - Data Collection
  - Data Analysis
  - Limitations
- Results
  - Dashboard
    - Power BI
    - Tableau
  - Maps
- Discussion
- Conclusion